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Roll No

ME-5002-CBGS

B.E. V Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Mechanical Measurement and Control

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) What do you mean by Calibration? Explain in brief the various types.
b) What are Primary, Secondary and tertiary measurements? Explain with suitable examples.
2. a) Explain the various sources of error. What is error propagation?
b) What is the importance of uncertainty analysis? How to carry out this?
3. a) Explain principle, construction and working of Resistance Temperature Detectors (RTD).
b) Explain any one type of differential pressure flow meter.
4. a) Discuss the construction and working of Whetstone's bridge circuit.
b) Explain the working of load cell with neat sketch.

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5.
 - a) Explain the difference between the open and closed loop control system with suitable examples.
 - b) Write brief note on modeling of fluid system.

6. Ten samples of a steel wire were tested on a universal testing machine. The breaking strengths in tones (t) of the samples were.
4.3 4.5, 4.7, 4.2, 4.5
4.6, 4.4, 4.6, 4.9, 4.5
Calculate the following
 - i) The mean value of the breaking strength
 - ii) Mean deviation of data
 - iii) Standard deviation
 - iv) Best estimate of precision of the apparatus and
 - v) Best estimate of the uncertainty in the data

7.
 - a) Explain the use of slip ring for measuring torque.
 - b) Discuss the construction and working of U tube manometer.

8. Write short notes of the following:
 - a) Repeatability and reproducibility
 - b) First and second order system
 - c) Transfer function
